MIL-S-3786/14E
03 June 1977
SUPERSEDING
MIL-S-3786/14D
18 December 1968

MILITARY SPECIFICATION SHEET

SWITCH, ROTARY, OPEN CONSTRUCTION, 1/2 AMPERE,
STYLE SR14

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the switch described herein shall consist of this document and the latest issue of specification MIL-S-3786.

INACTIVE FOR NEW DESIGN AFTER DATE OF THIS SPECIFICATION. USE MIL-S-3786/40.

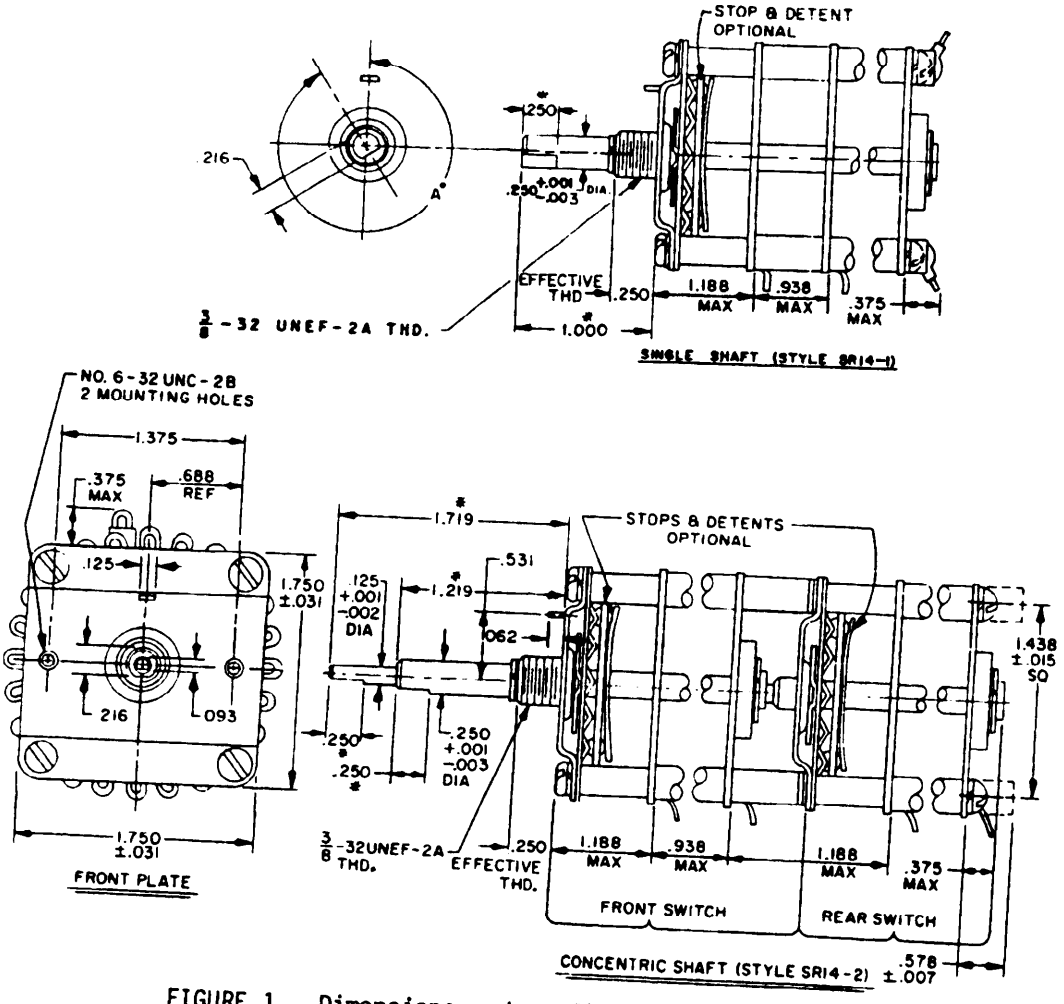


FIGURE 1. Dimensions and configuration.

FSC 5930

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INCHES	MM	INCHES	MM	INCHES	MM
.001	.03	.216	5.49	1.000	25.40
.002	.05	.250	6.35	1.188	30.18
.007	.18	.312	7.92	1.219	30.96
.015	.38	.375	9.53	1.375	34.93
.031	.79	.531	13.49	1.44	36.58
.062	1.57	.578	14.68	1.719	43.66
.093	2.36	.688	17.48	1.750	44.45
.125	3.18	.938	23.83	1.812	46.04

NOTES:
1. Dimensions are in inches.
2. Unless otherwise specified, tolerance is ±.015 (.38 mm).
3. The number of sections is optional.
4. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
5. Optional front plate of single piece construction, 1/8 thick aluminum with threaded inserts of steel for mounting holes.
6. For two hole mounting the bushing and non-turn device may be omitted.
7. S-construction supplied in bushing mounted switches only.
8. Sections are numbered consecutively starting with section No. 1 nearest its detent assembly.
* 9. For allowable variations, see MIL-S-3786 ordering data.
10. Shaft-flat angle A° is the angle between a line through the center of the shaft and center of the nonturn device and another line through the center of the shaft and perpendicular to the shaft flat.

FIGURE 1. Dimensions and configuration - Continued.

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REQUIREMENTS:

Dimensions and configuration: See figure 1.

Angle of throw: 11.25°, 15°, 22.5°, and 30°.

Terminals: The terminal tabs shall be bent at an angle of 0 ± 15 degrees from the plane of the section except that the common and corner terminals shall be bent at an angle of 45 ± 15 degrees from the plane of the section. When turret terminals are used, they shall be parallel to the shaft.

Material: The switch front plate and all metal parts which have wearing surfaces, shall be corrosion-resisting steel except see note 5 on figure 1 for optional front plate material.

Marking: All contact positions, including contacts not provided with terminals, shall be marked with consecutive numbers from 1 through 24 or 1 through 32, as applicable. Marking between terminals 1 and 2 will be provided on stators only. Contact number 1 location shall be the first possible assembly point of a contact in a counterclockwise direction from the top right-hand assembly screw.

Rear support: Switches which have more than five sections shall be supplied with four 6-32UNC-2B rear support spacers tapped to 1/4 inch minimum depth.

Rotational torque: The minimum and maximum values of torque determined for shaft rotation shall be within the limits shown in table I. Table I applies only to switches with one pole per deck.

Enclosure design: N or S.

TABLE I. Rotational-torque limits.

Construction	Temperature	Minimum	Torque (lb-in) 5 sections and less Maximum	Over 5 sections Maximum
N or S	Room	1-1/2	6	10
N	Minimum	1-1/2	8	13
S	Minimum	1-1/2	10	15

Torque (stops): Single shaft switches containing stops shall withstand a torque of 50 pound-inches applied to the shaft against the stops in clockwise and counterclockwise directions. Concentric shaft switches containing stops shall withstand a torque of 15 pound-inches applied to the .125-inch diameter shaft and 50 pound-inches applied to the .250 inch diameter shaft. The torque shall be applied in both a clockwise and counterclockwise direction. Single shaft switches with shaft seal containing stops shall withstand a torque of 25 pound-inches applied to the shaft against the stops in clockwise and counterclockwise direction.

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Life (rotational): The temperature-life characteristic shall be test condition E. After life (rotational), the torque shall change not more than 35 percent of its initial value. A cycle shall consist of 360 degrees minus the degrees in one throw in a clockwise direction and an equal number of degrees in a counterclockwise direction. The test loads for the applicable circuit conditions shall be as shown in table II. Each condition shall be switched by at least one rotor contact of the switch.

TABLE II. Electrical loads.

Environmental condition	Inductive load (2.8 Henries)		Resistive loads	
	Milliamperes	Volts, dc	Milliamperes	Volts
At atmospheric pressure	50	28	500 125 50	28 dc 115 Vrms 300 Vrms

Dielectric withstanding voltage: The applicable test voltage specified in table III shall be applied between the specified switch elements, except test voltage shall be reduced to 1,000 volts, rms after life test.

TABLE III. Dielectric-test voltages.

Altitude	Test voltage
	(Volts, rms)
At atmospheric pressure	1,500

Contact resistance: Contact resistance shall not exceed the following values:

Initial and after vibration and shock - 5 milliohms
After moisture-resistance and salt spray - 5 milliohms
After initial soak and completion of life test - 10 milliohms

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Army - MU, MI
Navy - AS, OS
Air Force - 11, 17, 99
DLA - ES

User activities:
Army - ME, AT, AV, SM
Navy - MC, CG
Air Force - 19

Preparing activity:
Air Force - 85

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-S-3786/14E Switch, Rotary, Open Construction, 1/2 Ampere, Style SR14 FSC 5930

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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B. RECOMMENDED WORDING CHANGE

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